A COUNTEREXAMPLE TO THE ERDŐS–MAHLER CONJECTURE

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Abstract. We disprove a 1939 conjecture of Erdős and Mahler by constructing, for every real $\tau \geq 4$, a Cantor set of numbers of irrationality exponent τ, each having infinitely many regular continued-fraction convergents of the form $2^a/3^b$.

1. Introduction

For an irrational real number ξ, its *irrationality exponent* $\mu(\xi)$ is the supremum of the real numbers $\nu > 0$ for which $0 < |\xi - u/v| < v^{-\nu}$ holds for infinitely many reduced fractions u/v with $v \geq 1$. Thus ξ is a Liouville number precisely when $\mu(\xi) = \infty$. Write p_n/q_n for the regular continued-fraction convergents of ξ, and $\mathrm{P}(m)$ for the largest prime factor of $|m|$ when m is a nonzero integer, with $\mathrm{P}(\pm 1) = 1$.

In 1939, Erdős and Mahler [3] investigated the prime factors of continued-fraction numerators and denominators. They proved that if $\mathrm{P}(q_{n-1}q_nq_{n+1})$ is bounded for infinitely many n, then ξ is a Liouville number. They conjectured that the same conclusion should hold if $\mathrm{P}(p_nq_n)$ is bounded for infinitely many n. The question was subsequently studied by Fraenkel [6] and recalled by Erdős in his reminiscences of Mahler [4]. In 2004, Bugeaud [1, Problem 43, p. 215] included the conjecture among the open problems on continued fractions.

Conjecture (Erdős–Mahler)**.** If $\mathrm{P}(p_nq_n)$ is bounded for infinitely many n, then ξ is a Liouville number.

We disprove the conjecture even when the selected numerators are powers of 2 and the corresponding denominators are powers of 3.

Theorem. *For every real $\tau \geq 4$, there exists a Cantor set $K_\tau \subset (1, 2)$ such that every $\xi \in K_\tau$ has irrationality exponent τ and infinitely many regular continued-fraction convergents of the form $2^a/3^b$, with a, b positive integers.*

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By Ridout's theorem [11], bounded prime factors of the denominators of infinitely many convergents already imply transcendence. Fraenkel [5, 6] studied rational approximation with prescribed prime factors and constructed Liouville numbers with infinitely many convergents subject to such restrictions. Fraenkel and Borosh [7] subsequently obtained non-Liouville examples by allowing additional cofactors in the numerators and denominators. When both cofactors are bounded, their Hausdorff-dimension formula gives zero. Lelis and Marques [8, Theorem 1.1] proved that ξ is a Liouville number if there is an increasing sequence (n_j) such that $\mathrm{P}(p_{n_j} q_{n_j})$ is bounded and $n_{j+1} - n_j = o(\log q_{n_j})$ as $j \to \infty$.

More recently, Bugeaud and Nguyen [2, Question 4.1, p. 32] revisited the conjecture and constructed further examples with restricted prime factors, leaving open whether the numbers they constructed are Liouville. This illustrates the main difficulty: prescribing infinitely many convergents does not exclude exceptionally good rational approximations at other denominators.

Our construction selects successive fractions $2^a/3^b$ in short intervals to the right of the preceding choice. Their prescribed displacement gives the lower bound $\mu(\xi) \geq \tau$; the difficulty is to exclude better rational approximations at other denominators. For $\nu > 4$, the neighborhoods of radius $v^{-\nu}$ around reduced fractions in $[1, 2]$ with denominator $v > D$ have a union of measure $O(D^{2-\nu})$. Proposition 2.1 gives the same upper bound, with an absolute implied constant, for the upper limiting proportion of prescribed fractions admitting such competing approximations. The estimate holds despite their rigid arithmetic structure and is independent of the intervals used in the construction.

The proof begins with an elementary counting estimate for irrational rotations, which handles small denominators. For the remaining approximations, the p-adic Subspace Theorem improves the elementary bound $\log v = O(N)$ to $\log v = o(N)$ on blocks of N indices. The main combinatorial observation is that, for a fixed sum of two indices, the product of the corresponding fractions takes at most two values. In denominator ranges $(V, V^\theta]$ with $1 < \theta < \nu/4$, these identities persist for sufficiently close rational approximants, whose product set consequently has cardinality $O(N)$. The Plünnecke–Ruzsa inequality then bounds the number of classes these approximants occupy modulo qth powers, for a suitable integer q. A lower bound for linear forms in logarithms, together with 3-adic valuations, confines the corresponding indices in each class to a short interval. The bound $\log v = o(N)$ makes the resulting estimates summable over all denominator ranges.

2. The counting proposition and the main theorem

We first state the counting proposition and deduce the main theorem from it. The proposition is proved in Section 3.

All logarithms are natural. We write $\mathbb{N} = \{1, 2, \dots\}$, take rational fractions with positive denominators, and use $|I|$ for the length of an interval or the cardinality of a finite set. Put

$$\alpha = \frac{\log 3}{\log 2}, \qquad a(b) = \lceil b\alpha \rceil, \qquad R(b) = \frac{2^{a(b)}}{3^b} \qquad (b \in \mathbb{N}).$$

Since α is irrational, $0 < a(b) - b\alpha < 1$. Thus $R(b)$ is the unique fraction in $(1, 2)$ with denominator 3^b and numerator a power of 2. It is in lowest terms. Let $\mathcal{D}$ be the set of denominators of the regular continued-fraction convergents of α.

At each stage of the construction, we shall select fractions $R(b)$ lying in a prescribed short interval. We must also exclude those admitting exceptionally close competing rational approximations. The following proposition controls these exceptions over the entire block $[N, 2N)$, independently of the target interval. Crucially, its constant is independent of the approximation exponent, which will approach 4 along the construction.

Fix $4 < \nu \leq 5$. A reduced fraction $u/v \in [1, 2]$ is *exceptional for b at exponent ν* if $u/v \neq R(b)$ and

$$\left| R(b) - \frac{u}{v} \right| < \frac{1}{16 v^\nu}. \tag{1}$$

For $D \geq 2$, let $E_N(D, \nu)$ count the integers $b \in [N, 2N)$ for which there is an exceptional fraction with $v > D$.

Proposition 2.1 (Exceptional approximations)**.** *There is an absolute constant C such that, for every $4 < \nu \leq 5$ and every $D \geq 2$,*

$$\limsup_{\substack{N \to \infty \\ N \in \mathcal{D}}} \frac{E_N(D, \nu)}{N} \leq C D^{2-\nu}.$$

We now fix a real $\tau \geq 4$ for the construction. For $t \in \mathbb{N}$ put $Q = 3^t$ and define

$$J_t = (R(t) + Q^{-\tau}, R(t) + 2Q^{-\tau}),$$
$$K(t) = [R(t) + Q^{-\tau}, R(t) + 3Q^{-\tau}].$$

The next fraction will be chosen in J_t, while $K(t)$ will contain the corresponding limit. Both intervals lie in $(1, 2)$ for sufficiently large t, since $R(t) \leq 2 - Q^{-1}$.

Put

$$D_t = Q^{\tau - 1}/6, \qquad \nu_t = 4 + 1/t.$$

An index b is *admissible at t* if $R(b) \in J_t$ and b has no exceptional fraction at exponent ν_t with denominator $v > D_t$. The threshold D_t separates the denominators handled by rational separation from those controlled by admissibility. For these parameters, Proposition 2.1 bounds the upper limiting proportion of exceptional indices by $O(Q^{-2(\tau-1)})$, whereas the proportion of indices b with $R(b) \in J_t$ tends to a quantity of order $Q^{-\tau}$. Thus sufficiently many admissible indices remain.

Lemma 2.2. *For every sufficiently large t, all sufficiently large $N \in \mathcal{D}$ admit two distinct indices $b, c \in [N, 2N)$ admissible at t.*

Proof. Fix t and put

$$\Phi(y) = 1 - \frac{\log y}{\log 2} \quad (1 \le y \le 2), \qquad \rho_t = |\Phi(J_t)|.$$

Write $\{x\} = x - \lfloor x \rfloor$ for fractional part. Since $\Phi(R(b)) = \{b\alpha\}$ and $|\Phi'(y)| = 1/(y \log 2)$, we have

$$\frac{Q^{-\tau}}{2 \log 2} \le \rho_t \le \frac{Q^{-\tau}}{\log 2}.$$

By the grid estimate (3), proved at the beginning of Section 3, there are $N\rho_t + O(1)$ indices with $R(b) \in J_t$. Among these we discard at most $E_N(D_t, \nu_t)$, since this last count includes exceptional indices in the whole of $[N, 2N)$. Writing $A_t(N)$ for the number of admissible indices that remain, we obtain from Proposition 2.1

$$\liminf_{\substack{N \to \infty \\ N \in \mathcal{D}}} \frac{A_t(N)}{N} \ge \rho_t - C D_t^{2-\nu_t} \ge \rho_t - 36 C Q^{-2(\tau-1)}.$$

Here $D_t^{2-\nu_t} \le D_t^{-2}$, and the last error divided by ρ_t is $O(Q^{2-\tau})$. Hence the right-hand side exceeds $\rho_t/2$ for every sufficiently large t. With such a t fixed, $A_t(N)$ therefore tends to infinity. □

Proof of the theorem. For any t large enough for Lemma 2.2, take $N \in \mathcal{D}$ larger than $Q = 3^t$ and sufficiently large to provide two admissible indices $b < c$ in $[N, 2N)$. We verify that their intervals lie strictly inside $K(t)$ and are disjoint.

For $d = b$ or c, the left endpoint of $K(d)$ exceeds $R(t) + Q^{-\tau}$. At the other endpoint, $R(d) \in J_t$ and $3^d \ge 2Q$ give

$$R(d) + 3 \cdot 3^{-\tau d} < R(t) + (2 + 3 \cdot 2^{-\tau}) Q^{-\tau} < R(t) + 3Q^{-\tau}.$$

To see disjointness, use the fact that the two indices were taken from the same interval $[N, 2N)$, so that $c < 2b$. The distinct fractions $R(b)$ and $R(c)$ can both be written with denominator 3^c, and hence differ by at least $3^{-c} > 3^{-2b}$. Each of the two intervals lies within $3 \cdot 3^{-\tau b} < 3^{-2b}$ to the right of its corresponding fraction.

Starting at a sufficiently large t_0, repeat the selection in each interval, labeling the two choices 0 and 1 in order of their indices. An infinite binary sequence determines a nested chain of intervals $K(t_j)$. Write $Q_j = 3^{t_j}$ and $r_j = R(t_j)$ along this chain. Since $Q_{j+1} \ge 2Q_j$, every interval at level j has length at most $2Q_0^{-\tau} 2^{-\tau j}$. The chain therefore has a unique common point ξ. Sequences agreeing through level j give points in the same interval at that level, and sequences making different choices give distinct points by disjointness. Thus the map from $\{0,1\}^{\mathbb{N}}$ to its image is continuous and injective. By compactness it is a homeomorphism onto a Cantor set $K_\tau \subset (1, 2)$.

Fix any $\xi \in K_\tau$ and consider its chain. Since the next interval lies in the interior of the preceding one, we have

$$Q_j^{-\tau} < \xi - r_j < 3Q_j^{-\tau} \qquad (j \geq 0). \tag{2}$$

If $\xi = u/v$ were rational, its distance from r_j would be at least $1/(vQ_j)$, in contradiction with (2) for large j. Moreover, $3Q_j^{-\tau} < 1/(2Q_j^2)$, so Legendre's criterion [1, Theorem 1.8, p. 10] makes each r_j a regular convergent of ξ. This gives infinitely many convergents $2^{a(t_j)}/3^{t_j}$ since the denominators strictly increase. For every $s < \tau$, their errors are less than Q_j^{-s} for all sufficiently large j, which gives $\mu(\xi) \geq \tau$.

For the upper bound, let $u/v \in [1, 2]$ be reduced with $v \geq Q_0$. Choose the unique j for which $Q := Q_j \leq v < Q_{j+1} =: S$. For $v \leq D_{t_j}$ we compare u/v with r_j. If $u/v = r_j$, then $v = Q$ and (2) gives $|\xi - u/v| > v^{-\tau}$. Otherwise rational separation gives

$$|\xi - u/v| \geq \frac{1}{vQ} - \frac{3}{Q^\tau} \geq \frac{1}{2vQ} \geq \frac{1}{2v^2},$$

where we used $6v \leq Q^{\tau-1}$ and $Q \leq v$.

For $D_{t_j} < v < S$ we compare instead with r_{j+1}. The condition $v < S$ ensures that $u/v \neq r_{j+1}$, and gives

$$|r_{j+1} - u/v| \geq \frac{1}{Sv} > S^{-2}.$$

Since $v < S$ and $S \geq 3$, we have $6v < S^3$, and hence

$$|\xi - r_{j+1}| < \frac{3}{S^\tau} \leq \frac{3}{S^4} \leq \frac{1}{2Sv} \leq \frac{1}{2}|r_{j+1} - u/v|.$$

As $v > D_{t_j}$, the admissibility of t_{j+1} at t_j gives

$$|\xi - u/v| \geq \tfrac{1}{2}|r_{j+1} - u/v| \geq \frac{1}{32v^{\nu_{t_j}}}.$$

For $Q_j \leq v < Q_{j+1}$ and $u/v \neq r_j$, the two ranges give $|\xi - u/v| \geq (32v^{\nu_{t_j}})^{-1}$. Given $\varepsilon > 0$, we have $\nu_{t_j} \leq 4 + \varepsilon/2$ for all sufficiently large j. For all sufficiently large v we also have $v^{\varepsilon/2} \geq 32$, so $|\xi - u/v| \geq v^{-4-\varepsilon}$ for all such fractions. For the prescribed convergents themselves, (2) gives $|\xi - r_j| > Q_j^{-\tau}$. Since $\tau \geq 4$ and rationals outside $[1, 2]$ are bounded away from ξ, it follows that $\mu(\xi) \leq \tau$. Thus $\mu(\xi) = \tau$ for every $\xi \in K_\tau$. □

3. Proof of Proposition 2.1

We begin with the elementary rotation estimate used in Lemma 2.2. For every arc $I \subset \mathbb{R}/\mathbb{Z}$ and $N \in \mathcal{D}$,

$$\#\{N \leq b < 2N : \{b\alpha\} \in I\} = N|I| + O(1), \tag{3}$$

with an absolute implied constant. To prove this, let p/N be the convergent of α with denominator N. Since $(p, N) = 1$, the points

$$N\alpha + \frac{ip}{N} \pmod 1, \qquad 0 \leq i < N,$$

form a translate of the uniform grid of spacing $1/N$, which contains $N|I|+O(1)$ points in I. By [1, Corollary 1.4, p. 10], we have $|\alpha - p/N| < N^{-2}$. Thus replacing these points by $(N+i)\alpha$ moves each by less than $1/N$. Only grid points within this distance of an endpoint of I can change membership, and there are $O(1)$ such points.

Fix $4 < \nu \leq 5$ and let N tend to infinity through $\mathcal{D}$. We divide the exceptional approximations according as their denominators are at most $N^{1/3}$ or exceed it. In the first range we can sum the error term in (3) over all the fractions in question. In the second, the interval belonging to a single fraction has length smaller than the grid spacing $1/N$. Thus each fraction occurs for only a bounded number of indices. The problem in this range is to bound the number of distinct fractions that occur.

3.1. **The range of exceptional denominators.** For $V \geq 2$, there are $O(V^2)$ reduced fractions in $[1,2]$ with denominator in $(V, 2V]$. For such a fraction u/v, the image under Φ of its $(16v^\nu)^{-1}$-neighborhood in $[1,2]$ is an interval of length $O(V^{-\nu})$. The grid estimate (3) therefore gives $O(NV^{-\nu}+1)$ possible indices for this fraction. Summing first over the fractions and then over $V = 2^i D$, $i \geq 0$, with $V < N^{1/3}$, we bound the number of indices admitting an exceptional denominator in $(D, N^{1/3}]$ by an absolute constant times

$$\sum_V (NV^{2-\nu} + V^2) \ll ND^{2-\nu} + N^{2/3}. \tag{4}$$

The decreasing sum is bounded in terms of its first term, since $2^{2-\nu} \leq 1/4$. The increasing sum is $O(N^{2/3})$, so the accumulated grid error is $o(N)$.

For larger denominators, rational separation alone gives $16v^{\nu-1} < 3^b$ by (1). In particular it gives only $\log v = O(N)$ on $[N, 2N)$. To sum the estimates for separate denominator ranges we shall need $\log v = o(N)$ uniformly over the exceptional fractions. The following consequence of the p-adic Subspace Theorem supplies this improvement.

Lemma 3.1. *For every $\varepsilon > 0$, all sufficiently large b and all reduced $u/v \in [1,2]$ distinct from $R(b)$ satisfy*

$$|R(b) - u/v| \geq 3^{-\varepsilon b} v^{-2}.$$

Proof. Write $P = 2^{a(b)}$ and $S = 3^b$. If P/S is close to u/v, the integers Pv and Su are close relative to their size. We apply the Subspace Theorem to the primitive pair with this ratio. Put

$$d = \gcd(Pv, Su), \qquad U = Pv/d, \qquad W = Su/d, \qquad e = |Pv - Su|.$$

We have $e \geq 1$. Every prime divisor of d belongs to $\{2,3\}$, since any other common prime divisor would divide both u and v. For a positive integer n, put $[n]_6 = 2^{v_2(n)}3^{v_3(n)}$, where v_p is the usual p-adic valuation. With $|p|_p = p^{-1}$ we have

$$|UW|_2 |UW|_3 = [UW]_6^{-1} = \frac{d^2}{PS[uv]_6}.$$

At the real place we take the forms $X - Y, Y$, so the factor in the product there is $|(U - W)W| = eSu/d^2$. Taking X, Y at each of 2 and 3 gives

$$\Pi := |(U - W)W|\,|UW|_2|UW|_3 = \frac{eu}{P[uv]_6} \le 2v^2|P/S - u/v|.$$

For the last inequality, use $e = Sv|P/S - u/v|$, $u \le 2v$, and $S < P$.

Suppose now that the asserted inequality fails along a sequence $b \to \infty$. On this sequence

$$(Sv)^{-1} \le |P/S - u/v| < S^{-\varepsilon}v^{-2},$$

so $v < S$. It follows that $H = \max(U, W) \le 2Sv < 2S^2$ and hence

$$\Pi < 2S^{-\varepsilon} \le 2^{1+\varepsilon/2}H^{-\varepsilon/2}.$$

The forms at each of the three places have determinant one. The p-adic Subspace Theorem [12, p. 246, (1.3)–(1.5)] therefore implies that all such pairs of sufficiently large height lie on finitely many rational lines through the origin. Indeed, the right-hand side is smaller than $(U^2 + W^2)^{-\delta/2}$ for all sufficiently large H, with $\delta = \min(\varepsilon/4, 1/2) > 0$.

Each rational line contains at most one positive primitive pair, and there are only finitely many pairs of bounded height. Thus U/W belongs to a finite set not containing 1. On the other hand,

$$0 < |U/W - 1| = \frac{|P/S - u/v|}{u/v} < S^{-\varepsilon}v^{-2} \le 3^{-\varepsilon b} \longrightarrow 0$$

as $b \to \infty$, a contradiction. □

Let W_N be the largest value of $\log v$ among exceptional fractions for $N \le b < 2N$, taking $W_N = 0$ when there are none. The preceding rational separation bound ensures that the maximum is finite. For each $\varepsilon > 0$, Lemma 3.1 and (1) give $16v^{\nu-2} < 3^{\varepsilon b}$ as soon as N is sufficiently large. Since $b < 2N$,

$$\limsup_{N\to\infty,\ N\in\mathcal{D}} \frac{W_N}{N} \le \frac{2\varepsilon \log 3}{\nu - 2}.$$

Letting ε decrease to zero, we conclude that

$$W_N = o(N). \tag{5}$$

3.2. **Products in a fixed denominator range.** Choose θ with $1 < \theta < \nu/4$. The condition $4\theta < \nu$ ensures that the error in a product of two exceptional approximants is smaller than the separation between distinct rational products. For $w > 0$ put $H = e^w$ and $V = e^{w/\theta}$, so that $H = V^\theta$. Write $\mathcal{A}(w)$ for the set of indices $b \in [N, 2N)$ admitting an exceptional fraction with $v > N^{1/3}$ and $V < v \le H$. We shall first estimate $|\mathcal{A}(w)|$ for a single w, and only at the end choose the ranges over which to sum.

Assume $\mathcal{A}(w)$ is nonempty. For each of its indices choose one such fraction s_b, and let X be the set of distinct fractions so obtained. Write $M = |X|$. If the same fraction of denominator v is chosen several times, (3) bounds the

number of those indices by $O(1+Nv^{-\nu})=O(1)$, since $v>N^{1/3}$ and $\nu>4$. Consequently

$$|\mathcal{A}(w)| \ll M. \tag{6}$$

We may henceforth retain one index for each element of X.

For these indices, compare the products $s_b s_c$ with the fractions

$$R(b)R(c)=2^{a(b)+a(c)}3^{-(b+c)}.$$

For fixed $\ell=b+c$, the integer $a(b)+a(c)$ lies in $(\alpha\ell,\alpha\ell+2)$ and thus takes at most two values. There are fewer than $2N$ choices for ℓ. It follows that the products $R(b)R(c)$ take at most $4N$ values.

The choice of θ allows us to transfer this bound to $XX=\{ss' : s,s'\in X\}$. Indeed, (1) and the fact that all fractions lie in $[1,2]$ give

$$|s_b s_c - R(b)R(c)| \le 2|s_b-R(b)|+2|s_c-R(c)| < \tfrac{1}{4}V^{-\nu}.$$

If $R(b)R(c)=R(b')R(c')$, the two corresponding products of elements of X thus differ by less than $\frac{1}{2}V^{-\nu}$. Their reduced denominators are at most H^2, so distinct products would differ by at least H^{-4}. Since $H^4=V^{4\theta}<V^{\nu}$, the products are equal. We have proved

$$|XX| \le 4N. \tag{7}$$

When M is a substantial fraction of N, (7) bounds the ratio $|XX|/M$. We use this ratio to bound the number of classes occupied by X modulo rational qth powers. Two positive rationals belong to the same class when their quotient is a qth power in $\mathbb{Q}$. For finite subsets A,B of an abelian group, write $A\pm B=\{a\pm b : a\in A,\ b\in B\}$ and $m\cdot A=\{ma : a\in A\}$. In the next lemma the group law is written additively.

Lemma 3.2. *Let A be a finite nonempty subset of a torsion-free abelian group G, and put $L=|A+A|/|A|$. For $q=2^k$ with $k\ge 1$, the image of A in G/qG has at most L^{3k} elements.*

Proof. Choose one representative $a\in A$ of each class in the image of A. The sets $a-q\cdot A$ lie in distinct cosets of qG and each contains $|A|$ elements, since dilation by q is injective. Their union is contained in $A-q\cdot A$. It therefore suffices to bound $|A-q\cdot A|/|A|$.

For a single doubling, the Plünnecke–Ruzsa inequality [10, Theorem 1.2], applied to A and itself, gives

$$|A-2\cdot A| \le |A-A-A| \le L^3|A|.$$

For the iteration we use Ruzsa's triangle inequality $|U-W|\,|V| \le |U-V|\,|V-W|$ for finite nonempty sets [10, proof of Theorem 1.2, (3)]. Take $U=A$, $V=2^{i-1}\cdot A$, $W=2^i\cdot A$. Since $V-W=2^{i-1}\cdot(A-2\cdot A)$, injectivity of dilation gives

$$\frac{|A-2^i\cdot A|}{|A|} \le \frac{|A-2^{i-1}\cdot A|}{|A|}\,\frac{|A-2\cdot A|}{|A|} \qquad (i\ge 2).$$

Iterating the single-doubling bound proves the lemma. □

3.3. **The count at large denominators.** By Lemma 3.2 in $\mathbb{Q}_{>0}^{\times}$ and (7), the elements of X occupy at most $(4N/M)^{3k}$ classes modulo rational qth powers when $q = 2^k$. To bound the size of each class we choose k in terms of $x = \log(N/w)$. Here N is the length of the interval of indices and w is the upper bound on $\log v$ in the denominator range.

Lemma 3.3. *There are absolute constants $c, C, x_0 > 0$ such that, for $4 < \nu \le 5$, $1 < \theta < \nu/4$, $N \in \mathcal{D}$, and $w \ge 1$ with $x = \log(N/w) \ge x_0$,*

$$\frac{|\mathcal{A}(w)|}{N} \le C \exp\left(-\frac{cx}{\log(2+x)}\right).$$

Proof. We may suppose that $\mathcal{A}(w)$ is nonempty and $x \ge 2$. Let $q = 2^k$ be the least power of two with $q \ge x^2$, so that $x^2 \le q < 2x^2$. We shall show that within a class modulo qth powers the ratio of two exceptional fractions agrees with the ratio of the corresponding $R(b)$. The height of the first ratio will then bound the difference of the two indices.

Take distinct s_b, s_c in the same class and write $s_c/s_b = g^q$ with $g \in \mathbb{Q}_{>0}$, $g \ne 1$. For a positive rational r/s in lowest terms, write $h(r/s) = \log\max(r, s)$. Each of s_b, s_c has denominator at most H and numerator at most $2H$. Thus the numerator and denominator of their quotient in lowest terms are both at most $2H^2$. Taking the qth root gives

$$qh(g) = h(s_c/s_b) \le 2w + \log 2 \le 3w. \tag{8}$$

This gain of a factor q in the height is the reason for working within one power class.

Consider the difference of the logarithms of the two ratios,

$$\Lambda = (a(c) - a(b))\log 2 - (c - b)\log 3 - q\log g.$$

The logarithm has derivative at most one on $[1, 2]$. Comparing $\log s_b$ with $\log R(b)$, and likewise at c, we obtain from (1)

$$|\Lambda| < \tfrac{1}{8}V^{-\nu} = \tfrac{1}{8}\exp(-\nu w/\theta). \tag{9}$$

Suppose $\Lambda \ne 0$. We use the weighted coefficient parameter in Matveev's lower bound [9, Corollary 2.3, (2.6), with (1.3)]. For positive rational bases $\gamma_i \ne 1$ and integer coefficients m_i $(1 \le i \le 3)$, take

$$A_i \ge \max\{h(\gamma_i), |\log\gamma_i|, 0.16\}, \qquad B = \max\left\{1, \max_i \frac{|m_i|A_i}{A_3}\right\}.$$

For a nonzero form in their real logarithms the bound over $\mathbb{Q}$ is

$$\log\left|\sum_{i=1}^{3} m_i \log\gamma_i\right| > -C_0 A_1 A_2 A_3 (1 + \log B)$$

with an absolute C_0. By (8), the existence of distinct s_b, s_c in the same class forces $3w/q \ge h(g) \ge \log 2$. Thus we may take

$$A_1 = \log 2, \qquad A_2 = \log 3, \qquad A_3 = 3w/q,$$

since $|\log g| \le h(g)$. The cited bound requires only that the linear form be nonzero; no multiplicative independence of $2, 3$, and g is assumed.

Since $b, c \in [N, 2N)$, we have $|c - b| < N$ and $|a(c) - a(b)| \le \alpha N + 1$. Thus the coefficients of $\log 2$ and $\log 3$ are $O(N)$, whereas that of $\log g$ is q. With $A_3 = 3w/q$, the weighted definition of B gives $B \ll qN/w$ and hence $1 + \log B \ll 1 + \log q + x$. Substitution in Matveev's bound gives

$$-\log|\Lambda| \ll \frac{w}{q}(1 + \log q + x) \ll \frac{w}{x}.$$

This is where the choice $q \asymp x^2$ enters. On the other hand, (9) gives $-\log|\Lambda| > (\nu/\theta)w > 4w$. The two bounds are incompatible for x larger than an absolute constant. It follows that $\Lambda = 0$, as claimed.

The vanishing of Λ turns the rational-power relation into an exact identity between the prescribed fractions:

$$\frac{s_c}{s_b} = 2^{a(c)-a(b)}3^{b-c}.$$

Taking the valuation at 3 and using (8) gives $|b - c| \log 3 \le h(s_c/s_b) \le 2w + \log 2$. Once s_b is fixed, all the representative indices in its class consequently lie in an interval of length $O(w)$. They are distinct integers, so the class has $O(w)$ elements, with an absolute constant.

Multiplying by the bound for the number of classes gives

$$M \le C_1 w(4N/M)^{3k}.$$

To read this as a bound for the proportion of exceptional indices, put $y = M/(4N)$. Since $w/N = e^{-x}$, the preceding inequality gives $y^{3k+1} \le (C_1/4)e^{-x}$. Although q is of order x^2, only $k = \log_2 q \ll \log(2+x)$ enters this exponent. Taking the $(3k+1)$st root and applying (6) proves the assertion, after increasing x_0 if necessary. □

Completion of the proof of Proposition 2.1. It remains to sum the estimates over the denominator ranges. Their number need not be bounded; what matters is that (5) forces all nonempty ranges into the tail of a convergent series. Put

$$w_j = N\theta^{-j}, \qquad x_j = \log(N/w_j) = j\log\theta \qquad (j = 0, 1, 2, \ldots).$$

The intervals $(w_j/\theta, w_j]$ partition $(0, N]$. By (5), they contain the logarithms of all exceptional denominators exceeding $N^{1/3}$ once N is sufficiently large. For any j with $\mathcal{A}(w_j) \ne \varnothing$, an exceptional denominator in that range satisfies

$$\tfrac{1}{3}\log N < \log v \le w_j < \theta \log v \le \theta W_N.$$

Thus $w_j \ge 1$ for large N. Moreover, if $W_N > 0$, then $x_j > \log(N/(\theta W_N))$, which tends to infinity by (5). This places every nonempty range in the part of the estimate where Lemma 3.3 applies.

For fixed $\theta > 1$, the series

$$\sum_{j\geq 0} \exp\left(-\frac{cx_j}{\log(2+x_j)}\right)$$

converges, since for large j its terms are at most $\exp(-c_\theta\sqrt{j})$ for some $c_\theta > 0$. Given any tail of this series, the nonempty ranges all belong to that tail for sufficiently large N. Lemma 3.3 therefore gives $\sum_j |\mathcal{A}(w_j)| = o(N)$. An index may occur in several ranges, but this sum bounds the number admitting any exceptional denominator above $N^{1/3}$.

Combining with (4) yields

$$E_N(D,\nu) \leq C\big(ND^{2-\nu} + N^{2/3}\big) + o_\nu(N),$$

where C is absolute and θ has been fixed in terms of ν. Only the last term uses convergence for this particular θ. Dividing by N and taking the upper limit along $\mathcal{D}$ proves the proposition. □

AI use declaration

The author used ChatGPT and Codex in developing the counting argument of Section 3, including the reduction to twofold products and the use of rational-power classes, and in formulating the diagonal construction in Section 2. These tools were also used to review the exposition and check proofs and references. The author takes full responsibility for the mathematical arguments and citations.

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